

Michael Marchese

Toronto, Ontario

647-284-0570 | Michael.Marchese92@gmail.com | linkedin.com/in/michael-d-marchese

SUMMARY

Senior data leader and systems architect with deep expertise in AI-augmented workflow design, large-scale data integration, and building scalable data infrastructure across enterprise environments.

Led a 7-person distributed data team delivering mission-critical data systems across over 100 organizations across 4 product lines, with a consistent focus on automation, reusability, and intelligent tooling. Designed and deployed internal knowledge management systems powered by LLMs, established prompt engineering governance, and built AI-assisted production workflows that operate at scale. Combines strong SQL engineering foundations with a systems-design mindset — architecting metadata-driven frameworks and multi-tier data pipelines built for reliability, consistency, and growth.

PROFESSIONAL EXPERIENCE

VitalHub Corp.

Director, Reporting, Analytics & Technical Implementation (May 2026 – Present)

- Promoted to Director overseeing cross-functional reporting, analytics, and technical implementation strategy across over 100 organizations across 4 product lines

Manager, Data Integration & Reporting Solutions (May 2024 – May 2026)

AI Systems & Tooling

- Designed and deployed a structured internal knowledge base integrating LLM tooling to accelerate onboarding, eliminate repeated research effort, and standardize outputs across a 7-person team serving over 100 organizations across 4 product lines
- Built reusable AI-assisted workflows for SQL generation, reporting documentation, and client deliverable production — reducing per-task effort and improving consistency at scale
- Established prompt engineering standards and AI tooling governance ensuring team-wide adoption within a regulated healthcare context, balancing productivity gains with data sensitivity requirements
- Architected AI-augmented processes spanning the full delivery lifecycle: intake, analysis, SQL development, QA, and documentation — enabling junior team members to operate at higher capability levels

Data Architecture & Systems

- Architected a 3-tier migration model (metadata-driven staging, validation, and transformation layers) that reduced project timelines by 30% and standardized delivery across all client engagements
- Built metadata-driven SQL frameworks powering modular, reusable data structures (DEX_, SPEX_, LT_, TR_) across all client environments — enabling parameterized execution and rapid schema adaptation
- Delivered 17 full-cycle migrations including 6 multi-system consolidations, overseeing movement of 100M+ records across 15+ modules with zero critical data loss events
- Increased migration revenue by 40% through process efficiency and capacity gains driven by automation and reusable system design

Reporting & Analytics

- Delivered 100+ operational, ministry-level, and financial reports across healthcare domains
- Modernized legacy WebFOCUS reporting infrastructure and established scalable frameworks for ongoing delivery

Leadership

- Lead domestic and offshore reporting and data integration teams
- Speaker at VitalHub Case Management Summit on migration architecture and delivery methodology

Technical Business Analyst (Aug 2022 – May 2024)

- Subject matter expert for SQL engineering, data modeling, and reporting pipeline design
 - Built reusable SQL templates and reporting frameworks adopted as team standards
 - Led modernization efforts for both migration and reporting infrastructure
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Humber River Hospital

Senior Business Analyst (Apr 2021 – Aug 2022)

- Managed \$218M capital and operational portfolio
- Supported COVID-19 operations and workflow automation across clinical and administrative domains

Business Analyst (Dec 2020 – Apr 2021)

- Built financial and operational reporting models supporting executive decision-making
 - Identified \$1M+ in vendor refund opportunities through data analysis
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TD Bank

IT Analyst II (Jun 2018 – Dec 2020)

- Developed productivity dashboards and reporting automation tools for internal operations teams
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Scotiabank

Senior Financial Advisor (2015 – 2018)

- Consistently exceeded sales targets; led branch growth and client portfolio initiatives
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CORE COMPETENCIES

AI & Systems

- LLM Integration & Prompt Engineering
- AI-Augmented Workflow Design
- Knowledge Management Systems
- AI Governance in Regulated Environments
- Metadata-Driven Architecture
- Dynamic SQL Automation

Data Engineering & Infrastructure

- SQL Server (T-SQL) — Expert
- ETL / Data Pipeline Design
- Data Migration Architecture
- Kimball Methodology
- Microsoft Fabric / Azure Synapse
- Mirth Connect (HL7/FHIR integration)

Leadership & Delivery

- Distributed Team Management (domestic + offshore)
- Executive & Client Communication
- Cross-Functional Stakeholder Alignment
- Agile Project Delivery

EDUCATION

Bachelor of Commerce (Economics)
University of Toronto

TECHNICAL SKILLS

Category	Tools / Technologies
Databases	SQL Server, T-SQL, Azure Synapse / Microsoft Fabric
AI & Automation	LLM APIs, Prompt Engineering, AI Workflow Design
Reporting	WebFOCUS, Excel (advanced), Power BI
Integration	Mirth Connect (HL7/FHIR), ETL pipelines
Project Tools	Jira, Confluence, Azure DevOps
Cloud	AWS (foundational), Azure